Assignment 4

SYSC5108

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Given:

$$\boldsymbol{x}_t = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, \qquad \boldsymbol{h}_{t-1} = \begin{bmatrix} 0.05 \\ 0.2 \\ 0.15 \end{bmatrix}, \qquad \boldsymbol{b}_h = \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \end{bmatrix}, \qquad \boldsymbol{b}_y = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}$$

and the weight matrices:

$$W_{xh} = \begin{bmatrix} -0.07 & 0.05 \\ -0.04 & 0.10 \\ -0.05 & -0.05 \end{bmatrix}, \quad W_{hh} = \begin{bmatrix} -0.22 & -0.10 & 0.05 \\ -0.04 & -0.09 & -0.06 \\ -0.09 & 0 & -0.08 \end{bmatrix}, \quad W_{hy} = \begin{bmatrix} 0.06 & -0.01 & -0.18 \\ -0.06 & 0.13 & 0.14 \end{bmatrix}.$$

a) Hidden-layer pre-activation:

$$\boldsymbol{z}_h = W_{xh}\boldsymbol{x}_t + W_{hh}\boldsymbol{h}_{t-1} + \boldsymbol{b}_h.$$

First,

$$W_{xh}\boldsymbol{x}_t = \begin{bmatrix} -0.07(1) + 0.05(0.5) \\ -0.04(1) + 0.10(0.5) \\ -0.05(1) - 0.05(0.5) \end{bmatrix} = \begin{bmatrix} -0.045 \\ 0.010 \\ -0.075 \end{bmatrix},$$

and

$$W_{hh}\boldsymbol{h}_{t-1} = \begin{bmatrix} -0.22(0.05) - 0.10(0.2) + 0.05(0.15) \\ -0.04(0.05) - 0.09(0.2) - 0.06(0.15) \\ -0.09(0.05) + 0(0.2) - 0.08(0.15) \end{bmatrix} = \begin{bmatrix} -0.0235 \\ -0.0290 \\ -0.0165 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{z}_h = \begin{bmatrix} 0.0315 \\ 0.0810 \\ 0.0085 \end{bmatrix}.$$

Since the hidden units use ReLU,

$$\boldsymbol{h}_t = \mathrm{ReLU}(\boldsymbol{z}_h) = \begin{bmatrix} 0.0315 \\ 0.0810 \\ 0.0085 \end{bmatrix}.$$

Now compute the output pre-activation:

$$\boldsymbol{z}_y = W_{hy}\boldsymbol{h}_t + \boldsymbol{b}_y = \begin{bmatrix} 0.06 & -0.01 & -0.18 \\ -0.06 & 0.13 & 0.14 \end{bmatrix} \begin{bmatrix} 0.0315 \\ 0.0810 \\ 0.0085 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}.$$

Again using ReLU at the output,

$$\boldsymbol{y}_t = \mathrm{ReLU}(\boldsymbol{z}_y) = \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}.$$

Therefore the answer is:

$$\boxed{\boldsymbol{y}_t = \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}}$$

b) Let the target be

$$\boldsymbol{t} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Using squared error and the same convention as earlier assignments,

$$\delta_k = (y_k - t_k)\,\mathrm{ReLU}'(z_k).$$

Since all hidden and output pre-activations are positive, every ReLU derivative is 1. Thus the output-layer deltas are

$$\delta_6 = 0.09955 - 0 = 0.09955, \qquad \delta_7 = 0.10983 - 1 = -0.89017.$$

For the hidden layer,

$$\boldsymbol{\delta}_h = (W_{hy}^\top \boldsymbol{\delta}_y) \odot \mathrm{ReLU}'(\boldsymbol{z}_h).$$

So,

$$W_{hy}^\top \boldsymbol{\delta}_y = \begin{bmatrix} 0.06 & -0.06 \\ -0.01 & 0.13 \\ -0.18 & 0.14 \end{bmatrix} \begin{bmatrix} 0.09955 \\ -0.89017 \end{bmatrix} = \begin{bmatrix} 0.0593832 \\ -0.1167176 \\ -0.1425428 \end{bmatrix}.$$

Therefore the hidden-layer deltas are

$$\delta_3 = 0.0593832, \qquad \delta_4 = -0.1167176, \qquad \delta_5 = -0.1425428.$$

Hence the neuron deltas are

$$\boxed{\delta_3 = 0.05938,\ \delta_4 = -0.11672,\ \delta_5 = -0.14254,\ \delta_6 = 0.09955,\ \delta_7 = -0.89017.}$$

2

From Slide 10 of Chapter 6,

$$P = \begin{bmatrix} p & 1-p \\ 0 & 1 \end{bmatrix},$$

with the initial state at $n = 0$ equal to state 0, so

$$\boldsymbol{\pi}(0) = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

a) At time n,

$$\pi_0(n) = p^n, \qquad \pi_1(n) = 1 - p^n.$$

b) To revisit state 0, the chain must stay in state 0 in the first step. After reaching state 1, it never returns. Therefore,

$$\boxed{\Pr(\text{revisit state 0}) = p.}$$

c) For $0 \le p < 1$,

$$\bar{\pi}_0 = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} p^n = 0, \qquad \bar{\pi}_1 = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} (1-p^n) = 1.$$

Hence,

$$\boxed{\bar{\pi}_0 = 0, \qquad \bar{\pi}_1 = 1.}$$

d) Before the message is sent successfully, the sender can only be in state 0. Hence

$$\boxed{\Pr(\text{sender is in state } 0 \mid \text{message not yet sent}) = 1.}$$

3

The transition matrix is

$$P = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}.$$

a) With

$$Q_0 = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix},$$

the marginal distribution at time 1 is

$$Q_1 = PQ_0 = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 0.8 - 0.6p \\ 0.2 + 0.6p \end{bmatrix}.$$

b) Define

$$A = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad B = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Then

$$P = A + (2p-1)B.$$

Also,

$$A^2 = A, \qquad B^2 = B, \qquad AB = BA = 0.$$

Therefore,

$$P^{(n)} = (A + (2p-1)B)^n = A + (2p-1)^n B.$$

Substituting A and B,

$$P^{(n)} = \begin{bmatrix} \frac{1}{2} + \frac{(2p-1)^n}{2} & \frac{1}{2} - \frac{(2p-1)^n}{2} \\ \frac{1}{2} - \frac{(2p-1)^n}{2} & \frac{1}{2} + \frac{(2p-1)^n}{2} \end{bmatrix},$$

which is exactly the required result.

c) If $0 < p < 1$, then $|2p - 1| < 1$, so

$$\lim_{n\to\infty} (2p-1)^n = 0.$$

Hence the limiting probabilities are

$$\boxed{\lim_{n\to\infty} Q_n = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.}$$

Special cases:

$$p = 1 \Rightarrow P = I \text{ and } Q_n = Q_0, \qquad p = 0 \Rightarrow P^n \text{ oscillates and no limit exists.}$$

d) For $0 < p < 1$,

$$\boxed{\lim_{n\to\infty} P^{(n)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.}$$

For $p = 1$, the limit is I. For $p = 0$, the limit does not exist.

4

Let the original reward at time $t+1$ be R_{t+1}, and define a new reward

$$R'_{t+1} = R_{t+1} + c.$$

Under any policy π,

$$\begin{aligned} v'_\pi(s) &= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R'_{t+k+1} \,\middle|\, S_t = s\right] \\ &= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k (R_{t+k+1} + c) \,\middle|\, S_t = s\right] \\ &= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,\middle|\, S_t = s\right] + c\sum_{k=0}^{\infty} \gamma^k = v_\pi(s) + \frac{c}{1-\gamma}, \end{aligned}$$

for $|\gamma| < 1$.

Thus every state value is shifted by the same constant

$$\boxed{v_c = \frac{c}{1-\gamma}.}$$

Therefore,

$$v'_\pi(s_1) - v'_\pi(s_2) = (v_\pi(s_1) + v_c) - (v_\pi(s_2) + v_c) = v_\pi(s_1) - v_\pi(s_2).$$

So adding a constant to every reward does not change any relative ordering of states under any policy. Only the intervals between rewards matter for comparing policies.

5

Environment given:

State	Action	Next State	Probability	Reward
0	0	0	1	-1
0	1	1	1	1
1	0	1	1	1
1	1	0	1	-1

with $\gamma = 0.1$.

The greedy action is $a = 1$ in state 0, and $a = 0$ in state 1.

a) For two actions, an ε-greedy policy assigns

$$\Pr(\text{greedy action}) = 1 - \varepsilon + \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2}.$$

Given

$$\pi(1|0) = \pi(0|1) = 0.6,$$

we have

$$1 - \frac{\varepsilon}{2} = 0.6 \quad \Rightarrow \quad \varepsilon = 0.8.$$

b) Under the given policy,

$$V(0) = 0.4[-1 + 0.1V(0)] + 0.6[1 + 0.1V(1)],$$
$$V(1) = 0.6[1 + 0.1V(1)] + 0.4[-1 + 0.1V(0)].$$

By symmetry, $V(0) = V(1) = v$. Then

$$v = 0.4(-1 + 0.1v) + 0.6(1 + 0.1v) = 0.2 + 0.1v,$$

so

$$0.9v = 0.2 \quad \Rightarrow \quad v = \frac{2}{9} \approx 0.2222.$$

Therefore,

$$\boxed{V(0) = V(1) = \frac{2}{9} \approx 0.2222.}$$

Now compute the action values:

$$Q(0,0) = -1 + 0.1V(0) = -1 + \frac{1}{45} = -\frac{44}{45} \approx -0.9778,$$
$$Q(0,1) = 1 + 0.1V(1) = 1 + \frac{1}{45} = \frac{46}{45} \approx 1.0222,$$
$$Q(1,0) = 1 + 0.1V(1) = \frac{46}{45} \approx 1.0222,$$
$$Q(1,1) = -1 + 0.1V(0) = -\frac{44}{45} \approx -0.9778.$$

So

$$\boxed{Q(0,0) = -0.9778,\ Q(0,1) = 1.0222,\ Q(1,0) = 1.0222,\ Q(1,1) = -0.9778.}$$

c) The optimal Bellman equations are

$$V^*(0) = \max\{-1 + 0.1V^*(0),\ 1 + 0.1V^*(1)\},$$

$$V^*(1) = \max\{1 + 0.1V^*(1),\ -1 + 0.1V^*(0)\}.$$

Clearly the optimal actions are

$$\boxed{\pi^*(0) = 1, \qquad \pi^*(1) = 0.}$$

Under this policy,

$$V^*(1) = 1 + 0.1V^*(1) \quad \Rightarrow \quad V^*(1) = \frac{1}{0.9} = \frac{10}{9},$$

and

$$V^*(0) = 1 + 0.1V^*(1) = 1 + \frac{1}{9} = \frac{10}{9}.$$

Thus

$$\boxed{V^*(0) = V^*(1) = \frac{10}{9} \approx 1.1111.}$$

d) Using the given random numbers and the policy in part (a):

Trajectory starting from $s_0 = 0$:

$$0 \xrightarrow[a=0]{-1} 0 \xrightarrow[a=1]{1} 1 \xrightarrow[a=0]{1} 1.$$

This uses $0.94, 0.27, 0.30$.

Trajectory starting from $s_0 = 1$:

$$1 \xrightarrow[a=0]{1} 1 \xrightarrow[a=1]{-1} 0 \xrightarrow[a=1]{1} 1.$$

This uses $0.18, 0.75, 0.35$.

The returns are

$$G_0^{(1)} = -1 + 0.1(1) + 0.1^2(1) = -0.89, \qquad G_1^{(1)} = 1 + 0.1(1) = 1.1, \qquad G_2^{(1)} = 1,$$

$$G_0^{(2)} = 1 + 0.1(-1) + 0.1^2(1) = 0.91, \qquad G_1^{(2)} = -1 + 0.1(1) = -0.9, \qquad G_2^{(2)} = 1.$$

Using every-visit incremental Monte Carlo with $\alpha = 0.6$ and initial estimates 0:

$$\boxed{V(0) \approx 0.77856, \qquad V(1) \approx -0.22560.}$$

The action-value estimates are

$$\boxed{Q(0,0) = -0.534,\ Q(0,1) = 0.864,\ Q(1,0) = 0.786,\ Q(1,1) = -0.540.}$$

e) For off-policy TD control, use Q-learning:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right].$$

Starting from all zeros, and treating the trajectory endpoint at $t = 3$ as terminal for the last update, the final action values are

$$\boxed{Q(0,0) = -0.600,\ Q(0,1) = 0.840,\ Q(1,0) = 0.876,\ Q(1,1) = -0.564.}$$

f) For on-policy TD control, use SARSA:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma Q(s',a') - Q(s,a) \right].$$

Start with $Q(s,a) = 0$ for all pairs and the initial policy

$$\pi(1|0) = \pi(0|1) = 0.5.$$

Using the same random sequence from the start:

$$0.94,\ 0.27,\ 0.30,\ 0.18,\ 0.75,$$

the trajectory is

$$0 \xrightarrow[a=0]{-1} 0 \xrightarrow[a=1]{1} 1 \xrightarrow[a=0]{1} 1 \xrightarrow[a=0]{1} 1 \xrightarrow[a=1]{-1} 0.$$

Applying SARSA step-by-step gives

$$\boxed{Q(0,0) = -0.600,\ Q(0,1) = 0.600,\ Q(1,0) = 0.840,\ Q(1,1) = -0.600.}$$

6

The five 2D samples are

$$(2,-2), \quad (0,4), \quad (-3,-1), \quad (3,-3), \quad (-2,2).$$

Since the sample mean is

$$\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

the centered data matrix is the same as the original one:

$$X = \begin{bmatrix} 2 & 0 & -3 & 3 & -2 \\ -2 & 4 & -1 & -3 & 2 \end{bmatrix}.$$

The covariance matrix is

$$\Sigma = \frac{1}{5} X X^\top = \begin{bmatrix} 5.2 & -2.8 \\ -2.8 & 6.8 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_1 \approx 8.9120, \qquad \lambda_2 \approx 3.0880.$$

Therefore the one-dimensional PCA direction is the eigenvector corresponding to λ_1:

$$\boxed{\boldsymbol{u}_1 = \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix}.}$$

a) One-dimensional PCA:

$$\boxed{\boldsymbol{u}_1 = \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix}.}$$

b) The encoded 1D coordinates are

$$z_i = \boldsymbol{u}_1^\top \boldsymbol{x}_i.$$

Hence

$$\boxed{z = \begin{bmatrix} 2.8011 & -3.1934 & -1.0082 & 4.2016 & -2.8011 \end{bmatrix}.}$$

c) The reconstructed points are

$$\hat{\boldsymbol{x}}_i = \boldsymbol{\mu} + \boldsymbol{u}_1 z_i.$$

Therefore the reconstructed samples are approximately

$$\boxed{\begin{aligned} \hat{\boldsymbol{x}}_1 &= (1.6868,\ -2.2362), \\ \hat{\boldsymbol{x}}_2 &= (-1.9230,\ 2.5494), \\ \hat{\boldsymbol{x}}_3 &= (-0.6072,\ 0.8049), \\ \hat{\boldsymbol{x}}_4 &= (2.5302,\ -3.3544), \\ \hat{\boldsymbol{x}}_5 &= (-1.6868,\ 2.2362). \end{aligned}}$$

d) The percentage of variance explained is

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{8.9120}{8.9120 + 3.0880} \approx 0.7427.$$

Therefore,

$$\boxed{\text{explained variance ratio} \approx 74.27\%.}$$

7

We have the data

$$x = [1, 2, 3, 9, 10],$$

and the initial parameters

$$\pi_1 = 0.6, \qquad \pi_2 = 0.4, \qquad \mu_1 = 2.0, \qquad \mu_2 = 9.5,$$

with $\sigma_1^2 = \sigma_2^2 = 1$.

For one EM iteration:

E-step. The responsibilities are

$$\gamma_{i1} = \frac{\pi_1 \mathcal{N}(x_i \mid \mu_1, 1)}{\pi_1 \mathcal{N}(x_i \mid \mu_1, 1) + \pi_2 \mathcal{N}(x_i \mid \mu_2, 1)}, \qquad \gamma_{i2} = 1 - \gamma_{i1}.$$

Numerically,

x_i	γ_{i1}	γ_{i2}
1	1.0000	0.0000
2	1.0000	0.0000
3	1.0000	0.0000
9	0.0000	1.0000
10	0.0000	1.0000

to four decimal places.

$$N_1 = \sum_i \gamma_{i1} \approx 3, \qquad N_2 = \sum_i \gamma_{i2} \approx 2.$$

M-step.

$$\pi_1^{\text{new}} = \frac{N_1}{5} \approx 0.6, \qquad \pi_2^{\text{new}} = \frac{N_2}{5} \approx 0.4.$$

$$\mu_1^{\text{new}} = \frac{\sum_i \gamma_{i1} x_i}{N_1} \approx \frac{1+2+3}{3} = 2.0,$$
$$\mu_2^{\text{new}} = \frac{\sum_i \gamma_{i2} x_i}{N_2} \approx \frac{9+10}{2} = 9.5.$$

Therefore after one EM iteration,

$$\boxed{\pi_1^{\text{new}} \approx 0.6000, \quad \pi_2^{\text{new}} \approx 0.4000, \quad \mu_1^{\text{new}} \approx 2.0000, \quad \mu_2^{\text{new}} \approx 9.5000.}$$

This means that in practice the parameters remain essentially unchanged after one step because the two clusters are already well separated.

8

Interpret the listed joint probabilities in lexicographic order:

$$P(X_0, X_1) = \begin{bmatrix} P(0,0) \\ P(0,1) \\ P(1,0) \\ P(1,1) \end{bmatrix} = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{8} \\ \frac{1}{8} \end{bmatrix}.$$

Equivalently, the joint table is

	$X_1 = 0$	$X_1 = 1$
$X_0 = 0$	$\frac{1}{4}$	$\frac{1}{2}$
$X_0 = 1$	$\frac{1}{8}$	$\frac{1}{8}$

a) First compute the marginals:

$$P(X_1 = 0) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}, \qquad P(X_1 = 1) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8},$$

$$P(X_0 = 0) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}, \qquad P(X_0 = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Therefore

$$P(X_0 = 0 \mid X_1 = 0) = \frac{1/4}{3/8} = \frac{2}{3}, \qquad P(X_0 = 1 \mid X_1 = 0) = \frac{1}{3},$$

$$P(X_0 = 0 \mid X_1 = 1) = \frac{1/2}{5/8} = \frac{4}{5}, \qquad P(X_0 = 1 \mid X_1 = 1) = \frac{1}{5}.$$

So

$$\boxed{P(X_0 \mid X_1 = 0) = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}, \qquad P(X_0 \mid X_1 = 1) = \begin{bmatrix} \frac{4}{5} \\ \frac{1}{5} \end{bmatrix}.}$$

Similarly,

$$P(X_1 = 0 \mid X_0 = 0) = \frac{1/4}{3/4} = \frac{1}{3}, \qquad P(X_1 = 1 \mid X_0 = 0) = \frac{2}{3},$$

$$P(X_1 = 0 \mid X_0 = 1) = \frac{1/8}{1/4} = \frac{1}{2}, \qquad P(X_1 = 1 \mid X_0 = 1) = \frac{1}{2}.$$

Hence

$$\boxed{P(X_1 \mid X_0 = 0) = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}, \qquad P(X_1 \mid X_0 = 1) = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.}$$

b) Use the random numbers in order.

Initialization from independent uniform distributions:

$$r_1 = 0.94 \Rightarrow X_0^{(0)} = 1, \qquad r_2 = 0.27 \Rightarrow X_1^{(0)} = 0.$$

Now run Gibbs sampling for 3 iterations.

Iteration 1

$$r_3 = 0.30, \quad X_1^{(0)} = 0 \Rightarrow P(X_0 = 0 \mid X_1 = 0) = \frac{2}{3} \Rightarrow X_0^{(1)} = 0.$$

$$r_4 = 0.18, \quad X_0^{(1)} = 0 \Rightarrow P(X_1 = 0 \mid X_0 = 0) = \frac{1}{3} \Rightarrow X_1^{(1)} = 0.$$

Iteration 2

$$r_5 = 0.75, \quad X_1^{(1)} = 0 \Rightarrow X_0^{(2)} = 1,$$

because $0.75 > \frac{2}{3}$.

$$r_6 = 0.35, \quad X_0^{(2)} = 1 \Rightarrow X_1^{(2)} = 0,$$

because $0.35 \leq \frac{1}{2}$.

Iteration 3

$$r_7 = 0.61, \quad X_1^{(2)} = 0 \Rightarrow X_0^{(3)} = 0,$$

because $0.61 \leq \frac{2}{3}$.

$$r_8 = 0.42, \quad X_0^{(3)} = 0 \Rightarrow X_1^{(3)} = 1,$$

because $0.42 > \frac{1}{3}$.

So the Gibbs samples are

$$\boxed{(X_0^{(0)}, X_1^{(0)}) = (1,0), \quad (X_0^{(1)}, X_1^{(1)}) = (0,0), \quad (X_0^{(2)}, X_1^{(2)}) = (1,0), \quad (X_0^{(3)}, X_1^{(3)}) = (0,1).}$$